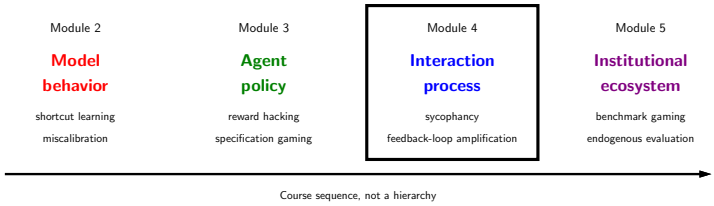
Lecture 11: Human Feedback, Preferences, and Reward Models

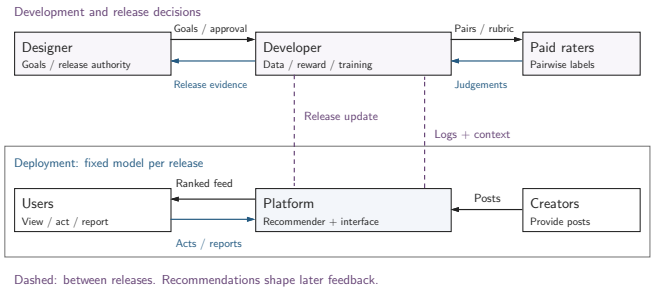
CS120: Introduction to AI Safety - October 27, 2026, Zachary Robertson

From agent policies to interaction loops



Actions change the environment. Interactions with users also shape preferences and future feedback.

Human participation is not effective control



Dashed: between releases. Recommendations shape later feedback.

Designers, evaluators, and end users need not be the same people.



Roadmap

What does preference feedback represent?

How is feedback produced?

Interaction loops and effective control

A recorded choice is not the preference itself

Stated wants

What does the user say they want?

Observed behavior

What does the user choose or watch?

Both depend on context, available alternatives, and how the question is asked.

Lambert, 2025

A lottery is a distribution over outcomes

Let $\mathcal{O}$ be a finite set of outcomes.

$$p_L(o) = \text{probability that lottery } L \text{ produces } o$$

$$p_L(o) \geq 0, \qquad \sum_{o \in \mathcal{O}} p_L(o) = 1$$

A pure outcome is a lottery with probability one on that outcome.

Bradley-Terry models choice between pure outcomes

$$P_\theta(\text{select } a \mid a, b) = \sigma(r_\theta(a) - r_\theta(b))$$

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Equal scores give selection probability $1/2$.

This models observed choices between pure lotteries.

Bradley-Terry is a utility model. Testing if it represents agent preference under risk (VNM) is a separate question.

Expected utility requires assumptions

Completeness: every comparison resolves, allowing ties.

Transitivity: comparisons are consistent across lotteries.

Continuity: strict comparisons survive small probability changes.

Independence: mixing in the same lottery preserves the comparison.

$$L \succeq M \iff \alpha L + (1-\alpha)N \succeq \alpha M + (1-\alpha)N$$

$$0 < \alpha < 1$$

Von Neumann Morgenstern axioms

Under these assumptions, there exists $u : \mathcal{O} \to \mathbb{R}$ such that

$$L \succeq M \iff \sum_{o\in\mathcal{O}} p_L(o)u(o) \geq \sum_{o\in\mathcal{O}} p_M(o)u(o)$$

The representation is unique up to a positive affine transformation:

$$u' = au + b, \qquad a > 0$$

von Neumann and Morgenstern, 1947

Individual models do not resolve aggregation

Utility scales are not automatically comparable across people.

Individual judgements → Aggregation rule → Shared model behavior

Whose judgements should shape the model?

Conitzer et al., 2024



Roadmap

What does preference feedback represent?

How is feedback produced?

Interaction loops and effective control

The interface can change the label

Compare

Choose between two outputs.

Rank

Order several outputs.

Rate

Score one output.

Option order, framing, and the ability to tie or abstain affect reports.

Lambert, 2025

Truthful revelation need not be feasible

Revelation principle: under its assumptions, the same equilibrium outcomes can be obtained through a truthful direct mechanism.

Indirect mechanism → Truthful direct mechanism

Comparing two outputs does not mean being able to specify preferences over every possible output.

Conitzer and Sandholm, 2004

Expressing intent is not verifying an output

Intent

Know what you want.

Instruction

Articulate it as a specification.

Verification

Check whether the output satisfies it.

Instructions, output observation, and verification are not interchangeable.

Expertise is task-specific

Code

Detect a security issue.

Art

Judge a subtle aesthetic difference.

Hiring an expert does not specify which judgement they should make.

Contracts shape what labels mean

A data worker may be asked to:

Express a personal preference.

Demonstrate how to complete a task.

Apply a rubric.

Recruitment, instructions, quality assurance, and payment shape the task.

Voluntary feedback comes from participants

Behavioral logs

Which videos are watched or liked?

Explicit comparisons

Which model response is selected?

Chatbot Arena offers model access in exchange for voluntary comparisons.

Chiang et al., 2024

Retain how the data was produced

A label alone does not preserve its collection conditions.

People and conditions

Recruitment, expertise, payment

Task and interface

Instructions, rubrics, alternatives

Selection

Filtering and quality assurance

Intended use

Where the judgements should apply

Protect workers, not only label quality

Data work can expose people to monotonous, unsettling, or toxic material.

Hazardous working conditions require worker protection, not only better labels.

Perrigo, TIME, 2023



Roadmap

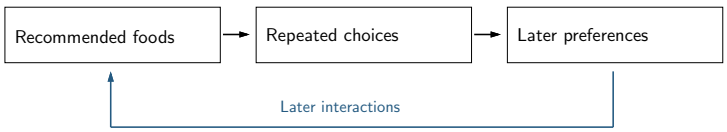
What does preference feedback represent?

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Preferences can change through interaction

Candy today; health and spending constraints later.



Recommendations also change what is explored and repeatedly chosen.

Jiang et al., 2019

A working demo can hide a safety failure

Repair a visible UI bug → Approve a backend patch → Visible workflow succeeds

The patch also removes an authorization check.

Approval can be an unsafe control action

Unsafe control action

Approve using only a partial view.

Hazard

Information is accessible to unauthorized users.

Harm: information is disclosed to unauthorized users.

Leveson and Thomas, 2018

Safety constraints need more than approval

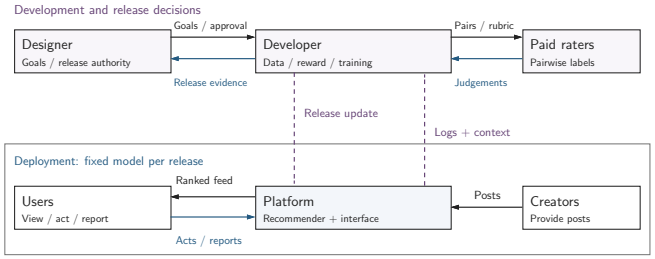
Scope edits: limit what the assistant can change.

Test separately: verify safety-critical functions, including authorization.

Retain rollback: restore a previous version after a failure.

Rollback can support recovery; it cannot undo information already disclosed.

Today's feedback shapes tomorrow's model



Dashed: between releases. Recommendations shape later feedback.

Outputs and ratings become data for a later release.

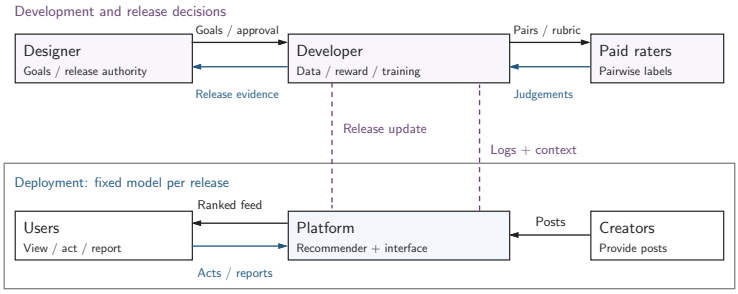
Observe, evaluate, and intervene

Observe	**Evaluate**	**Intervene**
What can the designer see?	Can they verify safety constraints?	Can they prevent or correct unsafe changes?

Human approval and feedback are not a fixed specification waiting to be articulated.

Perdomo et al., 2020

Social platform: control and feedback



Dashed: between releases. Recommendations shape later feedback.

Discussion: Human feedback on a social platform

A social platform aims to help users find worthwhile posts. A designer sets goals and approves releases; developers train a recommender using watch/like logs and paid pairwise judgements. Users and creators are not necessarily the raters.

Assume a fixed model between releases. The interface determines which posts and feedback options people see.

1. What does the feedback measure?

A user watches a post to the end but later says it was not worth seeing. What does each signal measure? Propose an elicitation change and a comparison that tests whether the interface changes the label.

Distinguish predicting a recorded choice from identifying a stable preference.

2. Whose judgements become data?

Contracted raters compare posts under a platform rubric. What must be recorded about recruitment, instructions, shown alternatives, and payment to interpret their labels? Identify one worker-protection requirement beyond label accuracy.

Explain why these judgements need not represent the end-user population.

3. What would justify rollout?

A new recommender scores better on paid-rater evaluations and passes its demo. But demo accounts have no blocked creators. Real users can block creators to keep their posts out of the feed. Would you approve rollout? What evidence would change your decision?

Where should the block rule be enforced? What might a passing test miss?

4. Feedback changes the next feedback

After a release, watch time and positive ratings rise. Could this happen without users finding the feed more worthwhile? Trace a loop through exposure, participation or preferences, and later training data.

What independently collected or longitudinal evidence would help distinguish explanations?